Design to Cost · AUDIT · TOOL DESIGN, DRAWINGS, SIMULATION, 3D CORRECTION

Is every task complete for every type of mould and tool?

Audit of release 2026-10-26 from a tool-room head's point of view. 70 of 70 task × tool-type cells are done for every die, mould and pattern type, including large parts of 10-20 kg on machines up to 3000-4000 t. The designer checks listed below are normal tool-room release checks. Every output was made from the sample parts and checked (pictures, drawings, STEP files, automated tests).

Task	Sheet metal: multi-stage dies	Sheet metal: progressive die	Plastic injection mould	Die-casting die	EPS mould	Stretch-blow mould	Foundry pattern / core box
3D assembly (STEP) with every part	**Done** Blanking, piercing, trimming, V-bending, U-bending (pressure pad, springs, gauge pins), deep drawing (punch, die, blank holder, cushion pins) and draw + trim: shoes, pillars, bushes, plates, punches, screws, dowels	**Done** Piercing, forming stations and cut-off at the pitch, stop pin, stripper, guides	**Done** Core / cavity × shrinkage, plates, ejection, cooling; sprue / cold runner + edge gates or hot-runner manifold + nozzles; slides with heel block and angle pin, lifters where undercuts are found	**Done** Mould structure + shot sleeve, biscuit, runners, overflows and vacuum vents	**Done** Cavity / core halves, steam chambers, back plates, fill injectors, steam vents	**Done** Cavity halves, neck and base inserts, blow pin	**Done** Match plate, cope / drag patterns with gating and riser pattern, core box, cope / drag flasks and flask pins
Every part as its own 3D (STEP)	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**
2D drawings: assembly + detail of every made part (PDF and DXF)	**Done** Assembly: iso, exploded, section, BOM. Plates: true views, hole table with fits and depths. Punches / dies: three views; profiles in DXF for wire cut	**Done** As multi-stage	**Done** Plates as above; core / cavity: three views + STEP (3D-machined)	**Done** As plastic	**Done** As plastic	**Done** As plastic	**Done** Patterns, core box and flasks: three views + STEP
Tolerances, fits, finish, heat treatment, labels on drawings	**Done** ISO 2768-mK, flatness / parallelism, fits (pillar g6, bush H7, dowel and punch m6/H7), finishes, hardness - all Design rules	**Done**	**Done** + insert-in-pocket fit, cavity polish	**Done**	**Done**	**Done**	**Done** Pattern tolerances and finish from Design rules
Logo and title block on every drawing (CAD and PDF)	**Done** Admin → Drawings: logo, company, projection, drawn / checked / approved, notes; DXF logo as hatches	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**
Simulation report (PDF)	**Done** Forming report: checks, blank and yield, stages, force, press, pictures, flat pattern; drawn parts detected	**Done** Same report	**Done** Wall / draft / sink / undercut maps, cycle, clamp, gate count, machine	**Done** Same analysis	**Done** EPS rules: draft, thin / thick walls, fill, steam, cooling, cycle	**Done** Blow rules: thin (over-stretched) / thick walls, blow, cooling, cycle	**Done** Foundry rules: pattern draft, misrun / hot spots, solidification, riser size
Simulation video (MP4)	**Done** Blanking, piercing, bends folding; failing bend in red	**Done**	**Done** Turntable of the wall, draft and risk maps with the result	**Done**	**Done**	**Done**	**Done**
Correct the 3D model in the software and release it as a new revision	**Done** Bend radius (outside size kept) and small holes; preview, STEP, new revision	**Done** Same	**Done** Draft added to the walls in 3D (default = min. draft + 0.5°); wall / rib changes still in CAD	**Done** As plastic	**Done** As plastic	**Done** As plastic	**Done** As plastic
Release pack ZIP for the tool room (3D + PDF + DXF + BOM)	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**

Task	Sheet metal: multi-stage dies	Sheet metal: progressive die	Plastic injection mould	Die-casting die	EPS mould	Stretch-blow mould	Foundry pattern / core box
Large parts 10-20 kg (presses / machines up to 3000-4000 t)	**Done** Presses to 3000 t, heavy die sets to 4800 × 2400, large sheet 4000 × 2000; sample tray 16.6 kg (draw + trim), channel 18.3 kg (U-bend)	**Done** As multi-stage	**Done** Machines to 4000 t, mould bases to 2200 mm, one cavity for large parts; sample tub 11.9 kg PP	**Done** HPDC to 2500 t, die holders to 1600 mm	**Done** Steam-chest base by size	**Done** -	**Done** Flask size from the part

Checked and right

- Blanking: die opening = blank size, punch = blank – clearance; piercing: punch = hole, die = hole + clearance (Design rules).
- Flat pattern unfolded from the 3D model; bend correction matches the designer's Rev B to 0.01 mm.
- Drawn parts (doubly curved walls) are recognised: blank → draw → trim, with draw die, blank holder and trim die in 3D.
- Large parts: plate / bed size of the press or machine is checked against the die set / mould base, not only the force.
- Mould: core and cavity with shrinkage; gate count and fill time from Design rules; slides and lifters placed where the analysis finds undercuts.
- Foundry: pattern equipment has a tool cost and a 3D design (match plate, patterns, gating / riser, core box, flasks).
- No number is fixed in the program: sizes, fits, tolerances and machine data are Master data (EXAMPLE values flagged until replaced).

Check before release (tool designer) - normal tool-room practice

- Press shut height, bed size and T-slots against the press data sheet.
- Slide / lifter face: the shape that forms the undercut is cut from the part, check the stroke and lock angle.
- V-bending: springback compensation by try-out; cooling layout of large moulds (generic straight channels).
- Foundry parting line is chosen from the largest section (heuristic): confirm with the foundry; EPS steam vent pattern size.
- Detail drawings give overall sizes; 3D surfaces and profiles come from the STEP / DXF (CNC, wire cut).

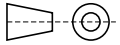
Not done (out of scope of the software)

- Wall-thickness and rib changes in 3D: done in CAD, the new revision is uploaded and checked again.
- Full FEM / flow simulation (Moldflow, AutoForm): the report is a rule-based check, use it before ordering a full analysis.

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master.

Design to Cost
Drawing no. **DTC-AUDIT-28**
Rev **-**

Scope	Roadmap stages 27-28	**Audited**	2026-10-26

General tolerance	ISO 2768-mK
Projection	
Scale	-
Sheet	1 / 1
Drawn	- · 2026-10-26
Checked	-
Approved	-